Esercitazione N° 1:
# La relazione fra peso corporeo e tasso metabolico basale

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## Creiamo il nostro profilo su Ubuntu

```
# questo è un commento!
# lanciamo il terminale per aggiungere un nuovo profilo
ctrl+alt+t
$ sudo adduser nome_battesimo -uid 663
$ pw di labx

# per eliminare un profilo utente (in caso di errore)
$ sudo deluser --remove-home nome_utente
$ pw di labx

# per recuperare la passwd dimenticata
# sostituire <user name> con il proprio nome utente
$ sudo passwd user name
$ pw di labx
$ nuova pw
```

**Creiamo il nostro profilo su Ubuntu**

# acquisiamo i privilegi dell’amministratore di sistema modificando il file di configurazione del Sistema Operativo (SO) `/etc/sudoers`

$ **`sudo visudo`**

# questo comando lancia l’editor di testo **nano** per creare ed aprire il file `/etc/sudoers.tmp` in modalità scrittura (per poterlo modificare); tale file è una copia di backup di `/etc/sudoers`

# Creiamo il nostro profilo su Ubuntu

```
  GNU nano 7.2                                   /etc/sudoers.tmp
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

                                   [ Lette 59 righe ]
^G Guida     ^O Salva      ^W Cerca        ^K Taglia   ^T Esegui      ^C Posizione  M-U Annulla  M-A Contrassegna M-] Parentesi  M-Q Precedente  ^B Indietro
^X Esci      ^R Inserisci  ^\ Sostituisci  ^U Incolla  ^J Giustifica ^/ Vai a riga M-E Ripeti   M-6 Copia        ^Q Cerca ind. M-W Successiva  ^F Avanti
```

# Creiamo il nostro profilo su Ubuntu

# Elenco degli amministratori di sistema
# portarsi con il cursore sull'ultima riga della lista
**alt-6** # copia
**ctrl-u** # incolla l'intera riga
# inserire il proprio nome utente
**ctrl-o** # per salvare
# cancellare l'estensione .tmp
**ctrl-x** # per chiudere **nano**
**alt-u** # in caso di errore
(per annullare l'ultimo comando inserito)

**Elenco degli amministratori di sistema**

```
GNU nano 7.2                                  /etc/sudoers.tmp
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

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# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
piero   ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d

^G Guida      ^O Salva       ^W Cerca        ^K Taglia    ^T Esegui      ^C Posizione   M-U Annulla  M-A Contrassegna M-] Parentesi  M-Q Precedente  ^B Indietro
^X Esci       ^R Inserisci   ^\ Sostituisci  ^U Incolla   ^J Giustifica  ^/ Vai a riga  M-E Ripeti   M-6 Copia        ^Q Cerca ind.  M-W Successiva  ^F Avanti
```
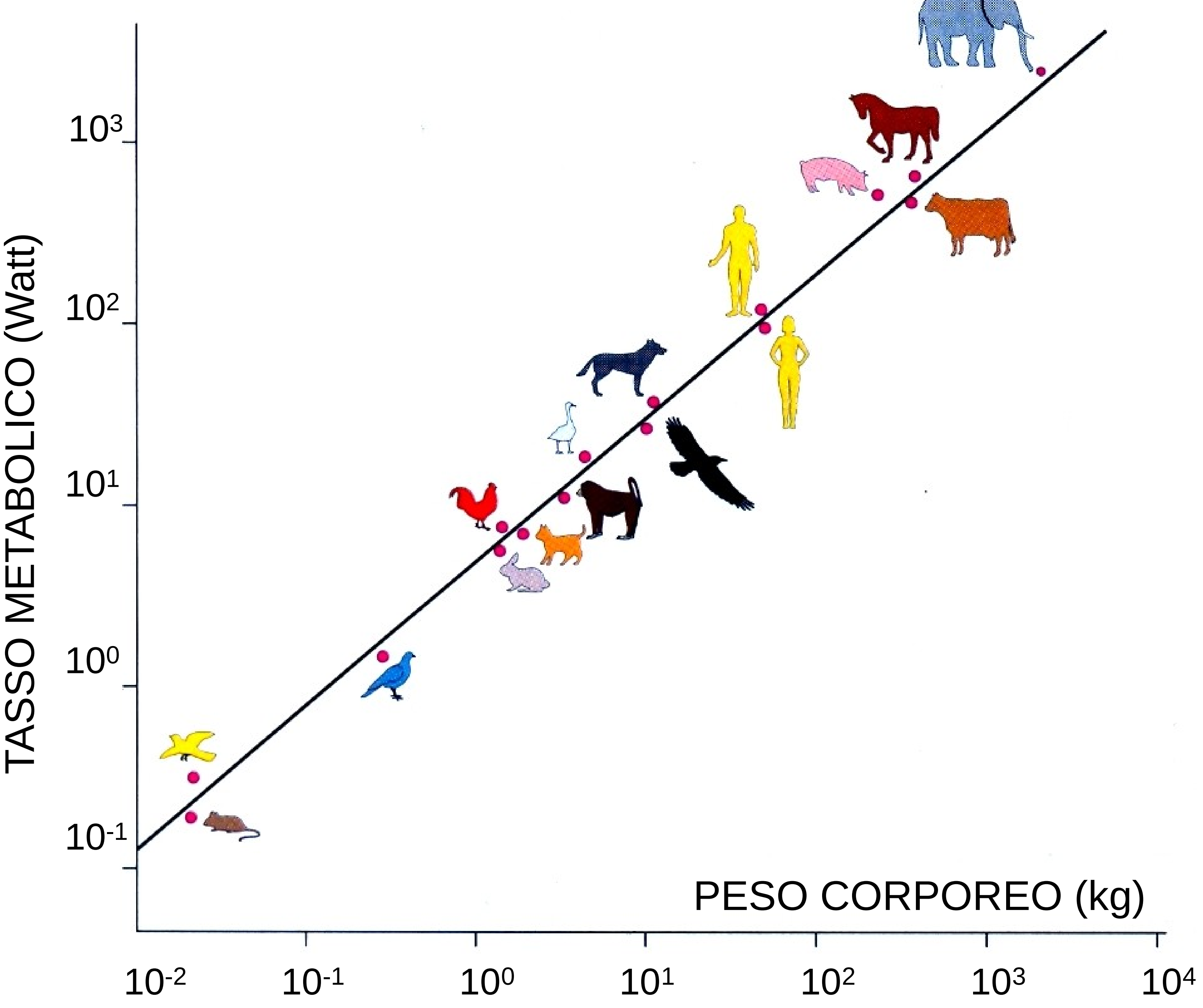
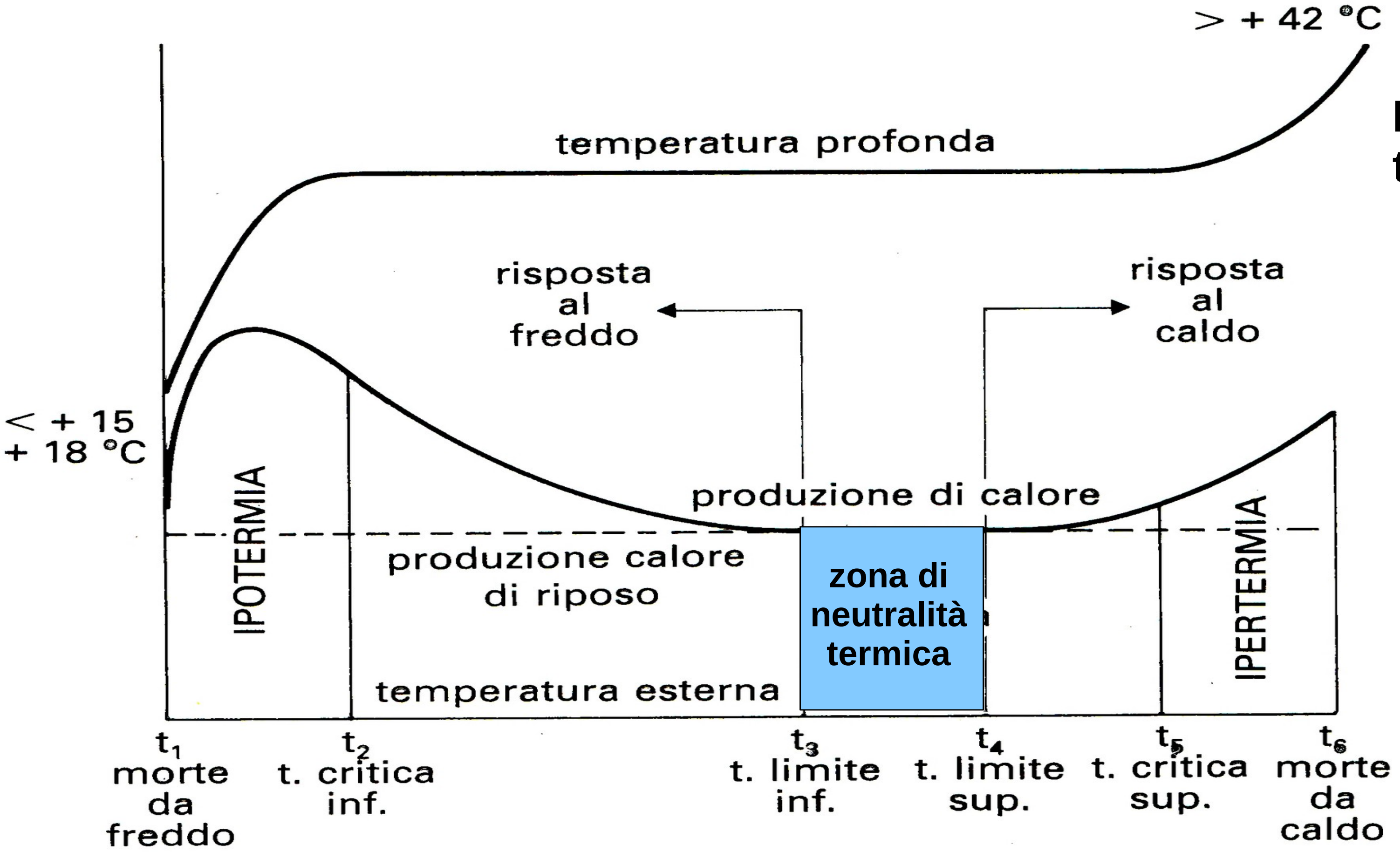

# La relazione fra peso corporeo e tasso metabolico basale

Andamento del tasso metabolico o Metabolismo Basale (MB) in watt (1 W = 1 J $*$ $s^{-1}$) in funzione del peso corporeo in Kg di varie specie animali. Sia i valori della variabile indipendente (il peso) sia quelli della variabile dipendente (il MB) sono espressi in scala logaritmica.

Da R.D. Martin, *Dimensioni del cervello ed evoluzione umana*, LE SCIENZE n. 319, marzo 1995, pag. 35-41.



## Metabolismo basale e termoregolazione




Produzione di calore (*termogenesi*) e andamento della *T* profonda in funzione della *T* esterna in un omeotermo: i valori di *T* limite inferiore e superiore sono espressi in °C e si riferiscono alla specie bovina, nella quale il limite inferiore della zona di neutralità termica è ≈ –5 °C. A *T* ambiente <, il ricambio energetico aumenta progressivamente: ad es. se la *T* esterna scende da –5 °C a –7,5 °C il dispendio energetico aumenta di 880 Kcal/die per il fabbisogno supplementare di calore necessario a mantenere costante la *T* corporea. Il limite superiore della zona di neutralità termica è ≈ 20 °C. A *T* ambientale compresa fra 20 e 25 °C il bovino conserva una *T* rettale costante (38÷39,5 °C) per qualsivoglia valore di umidità atmosferica, grazie ad efficaci meccanismi di termodispersione (vasodilatazione cutanea, lieve aumento della frequenza respiratoria). A *T* comprese fra +25°C e +40°C e con un'umidità del 50% la temperatura rettale inizia a salire: a +30°C e con umidità relativa > 70% il ricambio energetico cresce in modo rilevante. Sebbene i dati riportati siano suscettibili di variazione nel corso dei processi di acclimatazione, occorre sottolineare che la specie bovina (*Bos taurus*) è poco adattabile ai climi caldi. Da L.F. Giulio, RICAMBIO ENERGETICO E TERMOREGOLAZIONE DEGLI ANIMALI DOMESTICI, pag. 287-308. G. Aguccini, V. Beghelli e L.F. Giulio ed., *Fisiologia degli Animali Domestici con Elementi di Etologia*, UTET, Torino 1992.

# La relazione fra peso corporeo e tasso metabolico basale

Journal of Animal Ecology

BRITISH ECOLOGICAL SOCIETY

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## Scaling of basal metabolic rate with body mass and temperature in mammals

Andrew Clarke, Peter Rothery, Nick J. B. Isaac

First published: 22 March 2010 | https://doi.org/10.1111/j.1365-2656.2010.01672.x | Citations: 151

https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/j.1365-2656.2010.01672.x

https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/j.1365-2656.2010.01672.x

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## Acknowledgements

We thank Dr Kate Jones and the PanTheria team for the provision of data on ambient temperature, diet and trophic level for mammals. We also thank Pascale Perry and Laura Clark for extensive help with building and checking the databases.

## Supporting Information

**Appendix S1.** Data and taxonomy.

**Appendix S2.** Details of the statistical analysis.

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| Order | MSW3 | BMR & Bm | BMR, Bm & Tb | BMR, Bm, Tb & Ta |
|---|---|---|---|---|
| | **Number of species** | | | |
| Monotremata | 5 | 3 | 3 | 3 |
| Didelphimorphia | 87 | 11 | 11 | 11 |
| Paucituberculata | 6 | 0 | 0 | 0 |
| Microbiotheria | 1 | 0 | 0 | 0 |
| Notoryctemorphia | 2 | 1 | 1 | 1 |
| Dasyuromorpha | 71 | 22 | 21 | 18 |
| Peramelemorphia | 21 | 8 | 8 | 8 |
| Diprontodontia | 143 | 24 | 23 | 23 |
| Tubulidentata | 1 | 1 | 1 | 1 |
| Sirenia | 5 | 1 | 0 | 0 |
| Afrosoricida | 51 | 13 | 8 | 7 |
| Macroscelidea | 15 | 8 | 8 | 8 |
| Hyracoidea | 4 | 3 | 2 | 2 |
| Proboscidea | 3 | 0 | 0 | 0 |
| Cingulata | 21 | 9 | 9 | 9 |
| Pilosa | 10 | 6 | 6 | 6 |
| Scandentia | 20 | 3 | 2 | 2 |
| Dermoptera | 2 | 0 | 0 | 0 |
| Primates | 76 | 27 | 14 | 13 |
| Rodentia | 277 | 285 | 226 | 200 |
| Lagomorpha | 92 | 11 | 9 | 9 |
| Erinaceomorpha | 24 | 7 | 7 | 6 |
| Soricomorpha | 28 | 31 | 23 | 23 |
| Chiroptera | 116 | 81 | 68 | 65 |
| Pholidota | 8 | 5 | 5 | 5 |

Inserisci foglio...
Elimina foglio...
Rinomina foglio...
Duplica foglio
Sposta o copia foglio...
Seleziona tutti i fogli
Proteggi foglio...
Nascondi foglio
Visualizza linee della griglia
Colore linguetta...
Eventi del foglio...

Total By Order | Species With Codes | Species List

Foglio 1 di 3 PageStyle_Total By Order Inglese (UK) Media: ; Somma: 0 160%

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# Basal Metabolic Rate, Body Mass & Temperature in Mammals

## Data set for Clarke, Rothery & Isaac

Species names and Order from MSW3

Order shown in bold, lower case

Higher Groups used for taxonomic analysis shown in bold capitals

Ambient temperature (Ta) data kindly provided by PanTHERIA database team

| Species | BMR (W) | Bm (g) | Tb | Ta |
|---|---|---|---|---|
| **PROTOTHERIA: MONOTREMATA** | | | | |
| **Order: Monotremata** | | | | |
| *Tachyglossus aculeatus* | 2.404 | 2725 | 30.7 | 19.7 |
| *Zaglossus bruijni* | 6.778 | 10300 | 30.8 | 16.2 |
| *Ornithorhynchus anatinus* | 1.082 | 693 | 32.1 | 12.9 |
| **METATHERIA: MARSUPIALIA** | | | | |
| **Order: Didelphimorphia** | | | | |
| *Caluromys derbianus* | 1.255 | 329 | 35 | 19.3 |
| *Chironectes minimus* | 2.549 | 935 | 35 | 21.5 |
| *Didelphis marsupialis* | 3.185 | 1165 | 35 | 23.5 |
| *Didelphis virginiana* | 4.641 | 2488 | 35 | 15.4 |
| *Gracilinanus microtars* | 0.106 | 13 | 35 | 16.8 |
| *Lutreolina crassicaud* | 2.265 | 812 | 35.8 | 19.1 |
| *Marmosa robinsoni* | 0.547 | 122 | 34 | 19.3 |
| *Metachirus nudicauda* | 1.144 | 336 | 35 | 23.7 |
| *Monodelphis brevicau* | 0.318 | 75.5 | 33.7 | 24.4 |
| *Monodelphis domestic* | 0.335 | 104 | 32.6 | 21.9 |
| *Philander opossum* | 1.886 | 751 | 35.8 | 23.1 |

Species With Codes | Species List

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- ✓ Barra di stato
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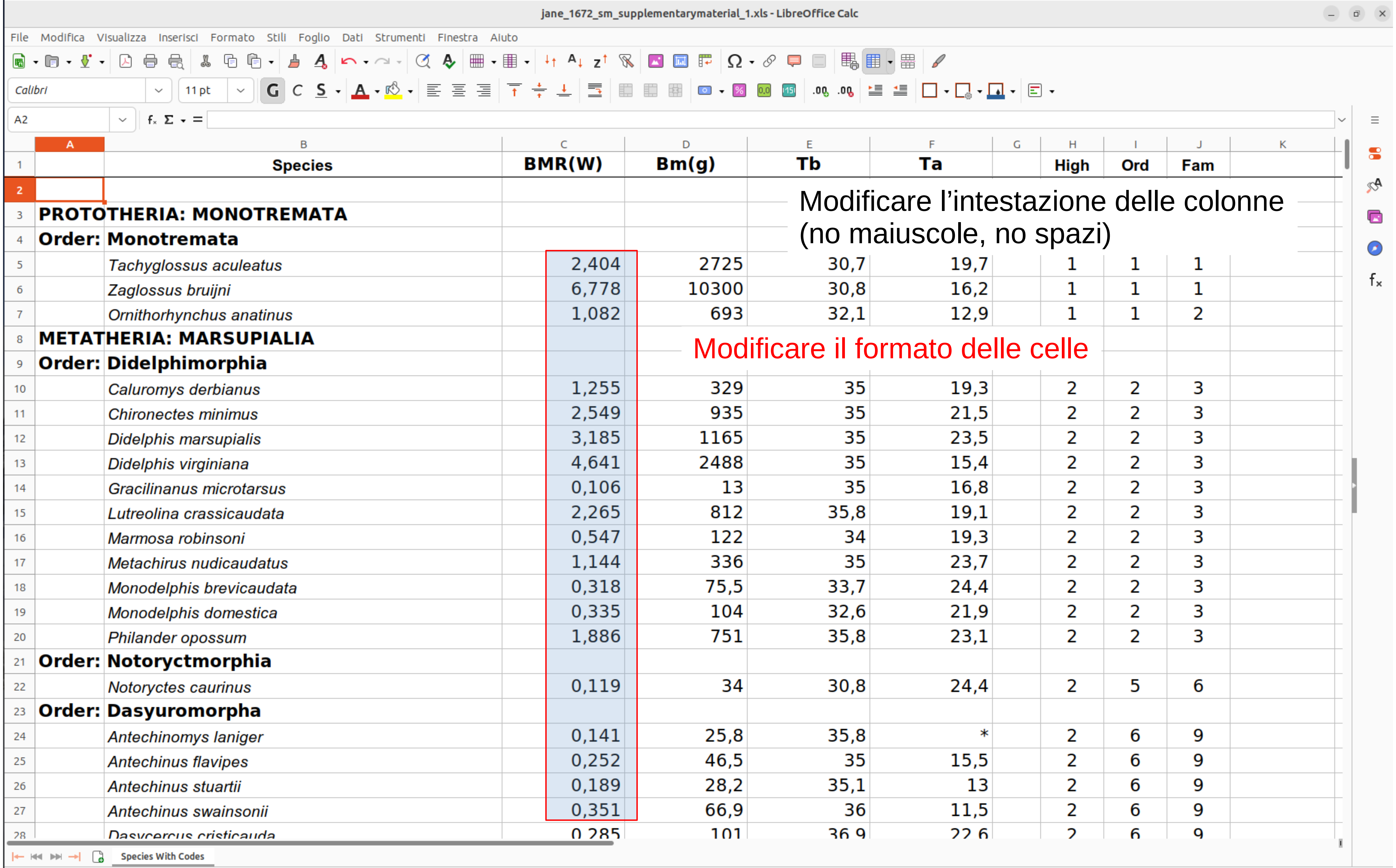
- ✓ Visualizza linee della griglia
- Griglia e linee guida
- Evidenzia valori Ctrl+F8
- Indicatore righe/colonne nascoste
- Mostra formula Ctrl+`
- Commenti
- Dividi finestra
- Blocca righe e colonne
- Blocca celle
- ✓ Barra laterale Ctrl+F5
- Stili F11
- Galleria
- Navigatore F5
- Elenco funzioni
- Schermo intero Maiusc+Ctrl+J
- Zoom

| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | **BMR(W)** | **Bm(g)** | **Tb** | **Ta** | | **High** | **Ord** | **Fam** | |
| 2 | | | | | | | | | | | |
| 3 | **PROT** | | | | | | | | | | |
| 4 | **Order** | | | | | | | | | | |
| 5 | | | 2,404 | 2725 | 30,7 | 19,7 | | 1 | 1 | 1 | |
| 6 | | | 6,778 | 10300 | 30,8 | 16,2 | | 1 | 1 | 1 | |
| 7 | | | 1,082 | 693 | 32,1 | 12,9 | | 1 | 1 | 2 | |
| 8 | **META** | | | | | | | | | | |
| 9 | **Order** | | | | | | | | | | |
| 10 | | | 1,255 | 329 | 35 | 19,3 | | 2 | 2 | 3 | |
| 11 | | | 2,549 | 935 | 35 | 21,5 | | 2 | 2 | 3 | |
| 12 | | | 3,185 | 1165 | 35 | 23,5 | | 2 | 2 | 3 | |
| 13 | | | 4,641 | 2488 | 35 | 15,4 | | 2 | 2 | 3 | |
| 14 | | *Gracilinanus microtarsus* | 0,106 | 13 | 35 | 16,8 | | 2 | 2 | 3 | |
| 15 | | *Lutreolina crassicaudata* | 2,265 | 812 | 35,8 | 19,1 | | 2 | 2 | 3 | |
| 16 | | *Marmosa robinsoni* | 0,547 | 122 | 34 | 19,3 | | 2 | 2 | 3 | |
| 17 | | *Metachirus nudicaudatus* | 1,144 | 336 | 35 | 23,7 | | 2 | 2 | 3 | |
| 18 | | *Monodelphis brevicaudata* | 0,318 | 75,5 | 33,7 | 24,4 | | 2 | 2 | 3 | |
| 19 | | *Monodelphis domestica* | 0,335 | 104 | 32,6 | 21,9 | | 2 | 2 | 3 | |
| 20 | | *Philander opossum* | 1,886 | 751 | 35,8 | 23,1 | | 2 | 2 | 3 | |
| 21 | **Order:** | **Notoryctmorphia** | | | | | | | | | |
| 22 | | *Notoryctes caurinus* | 0,119 | 34 | 30,8 | 24,4 | | 2 | 5 | 6 | |
| 23 | **Order:** | **Dasyuromorpha** | | | | | | | | | |
| 24 | | *Antechinomys laniger* | 0,141 | 25,8 | 35,8 | * | | 2 | 6 | 9 | |
| 25 | | *Antechinus flavipes* | 0,252 | 46,5 | 35 | 15,5 | | 2 | 6 | 9 | |
| 26 | | *Antechinus stuartii* | 0,189 | 28,2 | 35,1 | 13 | | 2 | 6 | 9 | |
| 27 | | *Antechinus swainsonii* | 0,351 | 66,9 | 36 | 11,5 | | 2 | 6 | 9 | |
| 28 | | *Dasycercus cristicauda* | 0,285 | 101 | 36,9 | 22,6 | | 2 | 6 | 9 | |

Species With Codes



| | Species | BMR(W) | Bm(g) | Tb | Ta | | High | Ord | Fam |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |
| **PROTOTHERIA: MONOTREMATA** | | | | | | | | | |
| **Order:** | **Monotremata** | | | | | | | | |
| | *Tachyglossus aculeatus* | 2,404 | 2725 | 30,7 | 19,7 | | 1 | 1 | 1 |
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| | *Ornithorhynchus anatinus* | 1,082 | 693 | 32,1 | 12,9 | | 1 | 1 | 2 |
| **METATHERIA: MARSUPIALIA** | | | | | | | | | |
| **Order:** | **Didelphimorphia** | | | | | | | | |
| | *Caluromys derbianus* | 1,255 | 329 | 35 | 19,3 | | 2 | 2 | 3 |
| | *Chironectes minimus* | 2,549 | 935 | 35 | 21,5 | | 2 | 2 | 3 |
| | *Didelphis marsupialis* | 3,185 | 1165 | 35 | 23,5 | | 2 | 2 | 3 |
| | *Didelphis virginiana* | 4,641 | 2488 | 35 | 15,4 | | 2 | 2 | 3 |
| | *Gracilinanus microtarsus* | 0,106 | 13 | 35 | 16,8 | | 2 | 2 | 3 |
| | *Lutreolina crassicaudata* | 2,265 | 812 | 35,8 | 19,1 | | 2 | 2 | 3 |
| | *Marmosa robinsoni* | 0,547 | 122 | 34 | 19,3 | | 2 | 2 | 3 |
| | *Metachirus nudicaudatus* | 1,144 | 336 | 35 | 23,7 | | 2 | 2 | 3 |
| | *Monodelphis brevicaudata* | 0,318 | 75,5 | 33,7 | 24,4 | | 2 | 2 | 3 |
| | *Monodelphis domestica* | 0,335 | 104 | 32,6 | 21,9 | | 2 | 2 | 3 |
| | *Philander opossum* | 1,886 | 751 | 35,8 | 23,1 | | 2 | 2 | 3 |
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| | *Notoryctes caurinus* | 0,119 | 34 | 30,8 | 24,4 | | 2 | 5 | 6 |
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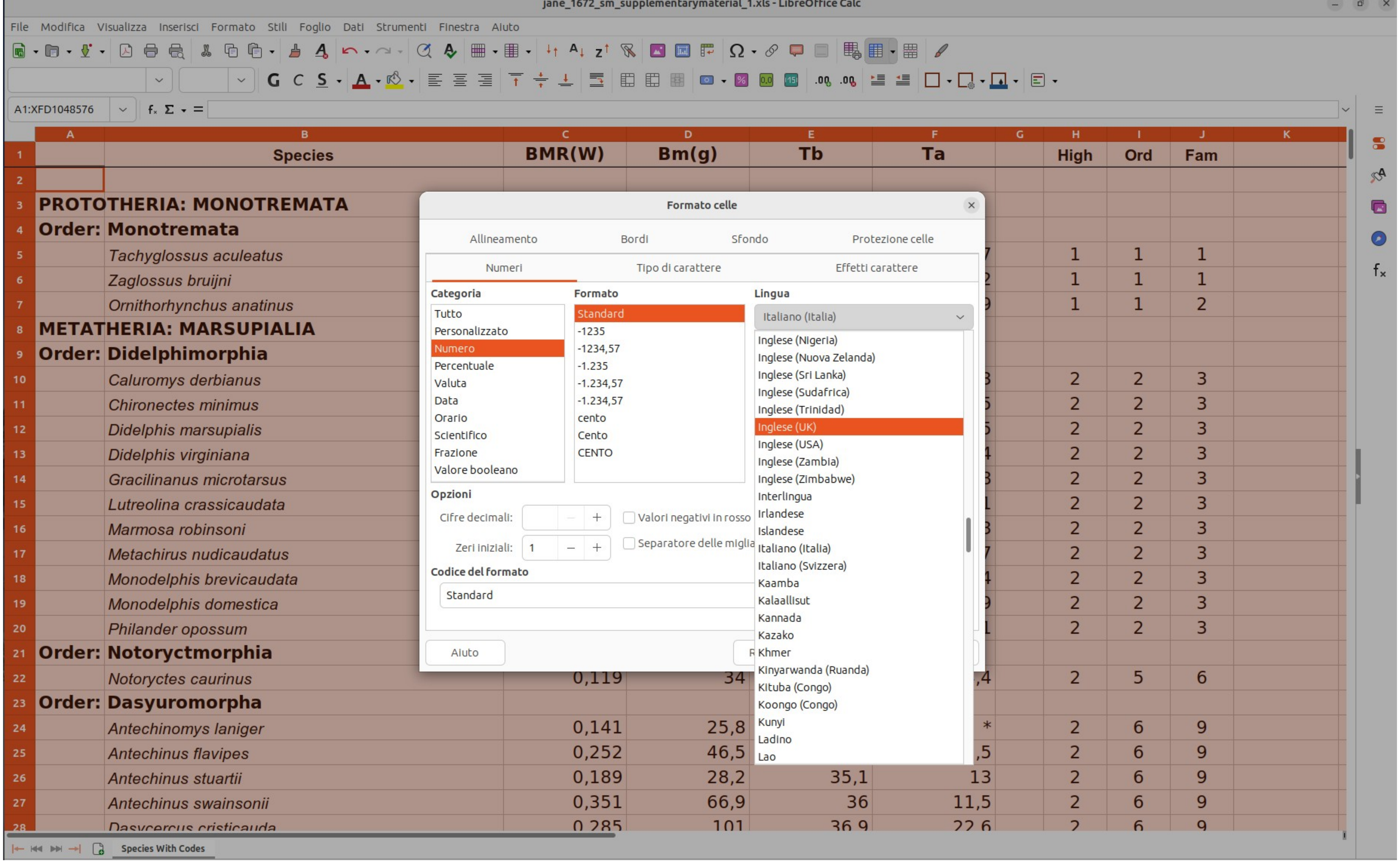
| | *Dasycercus cristicauda* | 0,285 | 101 | 36,9 | 22,6 | | 2 | 6 | 9 |

Modificare l'intestazione delle colonne (no maiuscole, no spazi)

Modificare il formato delle celle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Species
BMR(W)
Bm(g)
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Ta
High
Ord
Fam
PROTOTHERIA: MONOTREMATA
Order: Monotremata
Tachyglossus aculeatus
Zaglossus bruijni
Ornithorhynchus anatinus
METATHERIA: MARSUPIALIA
Order: Didelphimorphia
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Chironectes minimus
Didelphis marsupialis
Didelphis virginiana
Gracilinanus microtarsus
Lutreolina crassicaudata
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Monodelphis domestica
Philander opossum
Order: Notoryctmorphia
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0,119
34
Order: Dasyuromorpha
Antechinomys laniger
0,141
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28,2
35,1
13
Antechinus swainsonii
0,351
66,9
36
11,5
Dasycercus cristicauda
0,285
101
36,9
22,6
Formato celle
Allineamento
Bordi
Sfondo
Protezione celle
Numeri
Tipo di carattere
Effetti carattere
Categoria
Tutto
Personalizzato
Numero
Percentuale
Valuta
Data
Orario
Scientifico
Frazione
Valore booleano
Formato
Standard
-1235
-1234,57
-1.235
-1.234,57
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cento
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Inglese (Nigeria)
Inglese (Nuova Zelanda)
Inglese (Sri Lanka)
Inglese (Sudafrica)
Inglese (Trinidad)
Inglese (UK)
Inglese (USA)
Inglese (Zambia)
Inglese (Zimbabwe)
Interlingua
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Italiano (Italia)
Italiano (Svizzera)
Kaamba
Kalaallisut
Kannada
Kazako
Khmer
Kinyarwanda (Ruanda)
Kituba (Congo)
Koongo (Congo)
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Species With Codes

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| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | species | BasalMetabolicRate_W | BodyMass_g | Body_temp | Ambient_temp | | | | |
| 2 | | | | | | | | | | |
| 3 | PROTOTHERIA: MONOTREMATA | | | | | | | | | |
| 4 | Order: | Monotremata | | | | | | | | |
| 5 | | *Tachyglossus aculeatus* | 2.404 | 2725 | 30.7 | 19.7 | | 1 | | |
| 6 | | *Zaglossus bruijni* | 6.778 | 10300 | 30.8 | 16.2 | | 1 | | |
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| 8 | METATHERIA: MARSUPIALIA | | | | | | | | | |
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| 18 | | *Monodelphis brevicaudata* | 0.318 | 75.5 | 33.7 | 24.4 | | 2 | 2 | 3 |
| 19 | | *Monodelphis domestica* | 0.335 | 104 | 32.6 | 21.9 | | 2 | 2 | 3 |
| 20 | | *Philander opossum* | 1.886 | 751 | 35.8 | 23.1 | | 2 | 2 | 3 |
| 21 | Order: | Notoryctmorphia | | | | | | | | |
| 22 | | *Notoryctes caurinus* | 0.119 | 34 | 30.8 | 24.4 | | 2 | 5 | 6 |
| 23 | Order: | Dasyuromorpha | | | | | | | | |
| 24 | | *Antechinomys laniger* | 0.141 | 25.8 | 35.8 | * | | 2 | 6 | 9 |
| 25 | | *Antechinus flavipes* | 0.252 | 46.5 | 35 | 15.5 | | 2 | 6 | 9 |

Taglia
Copia
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Cancella formattazione diretta
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Sparkline
Formatta celle...

Species With Codes

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| Row | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| | | | **BasalMetabolicRate_W** | **BodyMass_g** | **Body_temp** | **Ambient_temp** |
| | | | 2.404 | 2725 | 30.7 | 19.7 |
| | | | 6.778 | 10300 | 30.8 | 16.2 |
| | | | 1.082 | 693 | 32.1 | 12.9 |
| | | | 1.255 | 329 | 35 | 19.3 |
| | | | 2.549 | 935 | 35 | 21.5 |
| | | | 3.185 | 1165 | 35 | 23.5 |
| | | | 4.641 | 2488 | 35 | 15.4 |
| 14 | | *Gracilinanus microtarsus* | 0.106 | 13 | 35 | 16.8 |
| 15 | | *Lutreolina crassicaudata* | 2.265 | 812 | 35.8 | 19.1 |
| 16 | | *Marmosa robinsoni* | 0.547 | 122 | 34 | 19.3 |
| 17 | | *Metachirus nudicaudatus* | 1.144 | 336 | 35 | 23.7 |
| 18 | | *Monodelphis brevicaudata* | 0.318 | 75.5 | 33.7 | 24.4 |
| 19 | | *Monodelphis domestica* | 0.335 | 104 | 32.6 | 21.9 |
| 20 | | *Philander opossum* | 1.886 | 751 | 35.8 | 23.1 |
| 21 | **Order:** | **Notoryctmorphia** | | | | |
| 22 | | *Notoryctes caurinus* | 0.119 | 34 | 30.8 | 24.4 |
| 23 | **Order:** | **Dasyuromorpha** | | | | |
| 24 | | *Antechinomys laniger* | 0.141 | 25.8 | 35.8 | * |
| 25 | | *Antechinus flavipes* | 0.252 | 46.5 | 35 | 15.5 |

Species With Codes

jane_1672_sm_supplementarymaterial_1.xls - LibreOffice Calc

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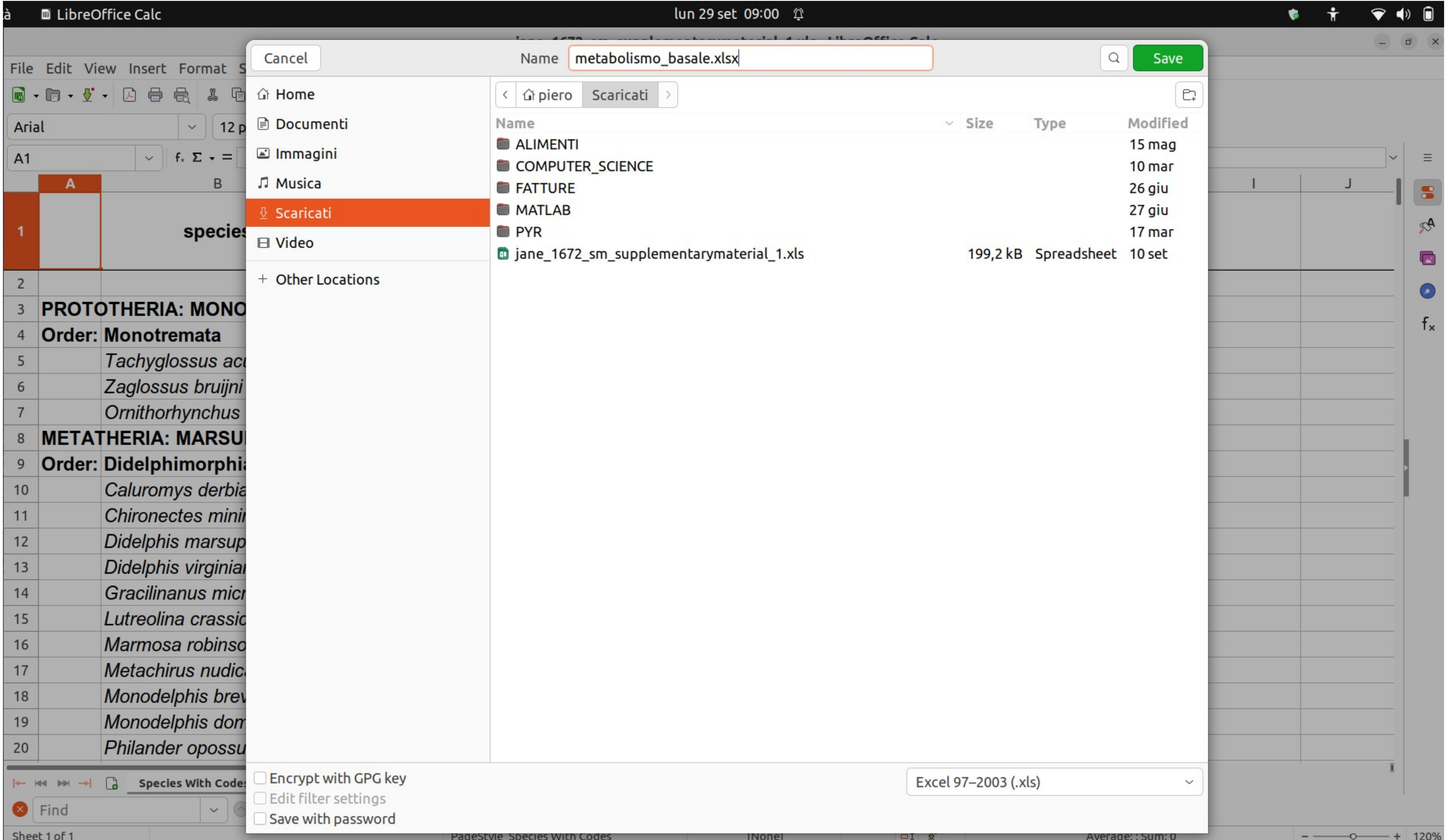
| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | | s | BasalMetabolicRate_W | BodyMass_g | Body_temp | Ambient_temp | |
| 2 | | | | | | | |
| 3 | **PROTOTHERIA: I EMATA** | | | | | | |
| 4 | **Order: Monotrem** | | | | | | |
| 5 | | *Tachyglos atus* | 2.404 | 2725 | 30.7 | 19.7 | |
| 6 | | *Zaglossus* | 6.778 | 10300 | 30.8 | 16.2 | |
| 7 | | *Ornithorhy atinus* | 1.082 | 693 | 32.1 | 12.9 | |
| 8 | **METATHERIA: M. LIA** | | | | | | |
| 9 | **Order: Didelphim** | | | | | | |
| 10 | | *Caluromy s* | 1.255 | 329 | 35 | 19.3 | |
| 11 | | *Chironect s* | 2.549 | 935 | 35 | 21.5 | |
| 12 | | *Didelphis s* | 3.185 | 1165 | 35 | 23.5 | |
| 13 | | *Didelphis* | 4.641 | 2488 | 35 | 15.4 | |
| 14 | | *Gracilinanus microtarsus* | 0.106 | 13 | 35 | 16.8 | |
| 15 | | *Lutreolina crassicaudata* | 2.265 | 812 | 35.8 | 19.1 | |
| 16 | | *Marmosa robinsoni* | 0.547 | 122 | 34 | 19.3 | |
| 17 | | *Metachirus nudicaudatus* | 1.144 | 336 | 35 | 23.7 | |
| 18 | | *Monodelphis brevicaudata* | 0.318 | 75.5 | 33.7 | 24.4 | |
| 19 | | *Monodelphis domestica* | 0.335 | 104 | 32.6 | 21.9 | |
| 20 | | *Philander opossum* | 1.886 | 751 | 35.8 | 23.1 | |
| 21 | **Order: Notoryctmorphia** | | | | | | |
| 22 | | *Notoryctes caurinus* | 0.119 | 34 | 30.8 | 24.4 | |
| 23 | **Order: Dasyuromorpha** | | | | | | |
| 24 | | *Antechinomys laniger* | 0.141 | 25.8 | 35.8 | * | |
| 25 | | *Antechinus flavipes* | 0.252 | 46.5 | 35 | 15.5 | |
| 26 | | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | |

Species With Codes

| | species | BasalMetabolicRate_W | BodyMass_g | Body_temp | Ambient_temp |
|---|---|---|---|---|---|
| | | | | | |
| **PROTOTHERIA: MONOTREMATA** | | | | | |
| **Order: Monotremata** | | | | | |
| | *Tachyglossus aculeatus* | 2.404 | 2725 | 30.7 | 19.7 |
| | *Zaglossus bruijni* | 6.778 | 10300 | 30.8 | 16.2 |
| | *Ornithorhynchus anatinus* | 1.082 | 693 | 32.1 | 12.9 |
| **METATHERIA: MARSUPIALIA** | | | | | |
| **Order: Didelphimorphia** | | | | | |
| | *Caluromys derbianus* | 1.255 | 329 | 35 | 19.3 |
| | *Chironectes minimus* | 2.549 | 935 | 35 | 21.5 |
| | *Didelphis marsupialis* | 3.185 | 1165 | 35 | 23.5 |
| | *Didelphis virginiana* | 4.641 | 2488 | 35 | 15.4 |
| | *Gracilinanus microtarsus* | 0.106 | 13 | 35 | 16.8 |
| | *Lutreolina crassicaudata* | 2.265 | 812 | 35.8 | 19.1 |
| | *Marmosa robinsoni* | 0.547 | 122 | 34 | 19.3 |
| | *Metachirus nudicaudatus* | 1.144 | 336 | 35 | 23.7 |
| | *Monodelphis brevicaudata* | 0.318 | 75.5 | 33.7 | 24.4 |
| | *Monodelphis domestica* | 0.335 | 104 | 32.6 | 21.9 |
| | *Philander opossum* | 1.886 | 751 | 35.8 | 23.1 |
| **Order: Notoryctmorphia** | | | | | |
| | *Notoryctes caurinus* | 0.119 | 34 | 30.8 | 24.4 |
| **Order: Dasyuromorpha** | | | | | |

Species With Codes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species
PROTOTHERIA: MONO
Order: Monotremata
Tachyglossus acu
Zaglossus bruijni
Ornithorhynchus
METATHERIA: MARSU
Order: Didelphimorphia
Caluromys derbia
Chironectes minir
Didelphis marsup
Didelphis virginiar
Gracilinanus micr
Lutreolina crassic
Marmosa robinso
Metachirus nudic
Monodelphis brev
Monodelphis dom
Philander opossu
Species With Code
Find
Sheet 1 of 1
Cancel
Name
metabolismo_basale.xlsx
Save
Home
Documenti
Immagini
Musica
Scaricati
Video
Other Locations
piero
Scaricati
Name
Size
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ALIMENTI 15 mag
COMPUTER_SCIENCE 10 mar
FATTURE 26 giu
MATLAB 27 giu
PYR 17 mar
jane_1672_sm_supplementarymaterial_1.xls 199,2 kB Spreadsheet 10 set
Encrypt with GPG key
Edit filter settings
Save with password
Excel 97–2003 (.xls)

# Installazione di tidyverse in R

ctrl+alt+t

\# apriamo il terminale

```
$ sudo apt install libssl-dev libcurl4-openssl-dev unixodbc-dev libxml2-dev libmariadb-dev libfontconfig1-dev libharfbuzz-dev libfribidi-dev libfreetype6-dev libpng-dev libtiff5-dev libjpeg-dev libwebp-dev
```

\# installiamo le dipendenze

\# per 4.5.1 (2025-06-13) -- "Great Square Root"

```
> install.packages('tidyverse')
```

\# installiamo <tidyverse>

# Installazione di altri pacchetti

ctrl+alt+t

```
$ R
> install.packages('readxl')
> install.packages('ggplot2')
> install.packages('ggthemes')
> library(readxl)
> library(ggthemes)
```

# La relazione fra peso corporeo e tasso metabolico basale

```
> library(readxl)
> BasalMetabolicRate =
read_excel('/home/piero/COMPUTER_SCIENCE/AI/metabolismo_basale.xlsx')
New names:
• `` -> `...1`
```

# ATTENZIONE: il path dovrà essere opportunamente modificato!

```
piero@piero-XPS-9320:~$ locate metabolismo_basale.xlsx
/home/piero/COMPUTER_SCIENCE/AI/metabolismo_basale.xlsx
/home/piero/COMPUTER_SCIENCE/DATA/metabolismo_basale.xlsx
/home/piero/MATERIALE_DIDATTICO_A.S._2025-2026/ESERCITAZIONI/
METABOLISMO_BASALE/metabolismo_basale.xlsx
...
piero@piero-XPS-9320:~$
```

# La relazione fra peso corporeo e tasso metabolico basale

```
> library(readxl)
> BasalMetabolicRate =
read_excel('/home/piero/COMPUTER_SCIENCE/AI/metabolismo_basale.xlsx')
New names:
• `` -> `...1`
> BasalMetabolicRate
# A tibble: 666 × 6
   ...1          species BasalMetabolicRate_W BodyMass_g Body_temp
Ambient_temp
   <chr>         <chr>                   <dbl>      <dbl> <chr>
<chr>
 1 PROTOTHERIA: … NA                     NA            NA NA         NA

 2 Order: Monotr… NA                     NA            NA NA         NA

 3 NA            Tachyg…                  2.40       2725 30.7
19.7
```

# La relazione fra peso corporeo e tasso metabolico basale

```
> BasalMetabolicRate$BasalMetabolicRate_W
 [1]     NA     NA  2.404  6.778  1.082     NA     NA  1.255
 2.549
 [10]  3.185  4.641  0.106  2.265  0.547  1.144  0.318  0.335
 1.886
 [19]     NA  0.119     NA  0.141  0.252  0.189  0.351  0.285
 0.46
...
```

# per estrarre una colonna

# La relazione fra peso corporeo e tasso metabolico basale

```
> is.na(BasalMetabolicRate$BasalMetabolicRate_W)
  [1]  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE
FALSE FALSE
 [13] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE
FALSE FALSE
 [25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
FALSE FALSE
...
```

# è un <NA>? (Not A number):
# se non è un numero (es. cella vuota) -> TRUE
# se è un numero -> FALSE

# La relazione fra peso corporeo e tasso metabolico basale

```
> !is.na(BasalMetabolicRate$BasalMetabolicRate_W)
  [1] FALSE FALSE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE
TRUE  TRUE
 [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE
TRUE  TRUE
 [25]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
TRUE  TRUE

...
```

# non è un <NA>? (Not A number):
# se è un numero (es. cella vuota) -> TRUE
# se non è un numero -> FALSE

# La relazione fra peso corporeo e tasso metabolico basale

```
> tasso_metabolico_basale = BasalMetabolicRate$BasalMetabolicRate_W[!is.na(BasalMetabolicRate$BasalMetabolicRate_W)]
> tasso_metabolico_basale
  [1]   2.404   6.778   1.082   1.255   2.549   3.185   4.641   0.106   2.265
 [10]   0.547   1.144   0.318   0.335   1.886   0.119   0.141   0.252   0.189
 [19]   0.351   0.285   0.462   3.172   1.356   3.281   2.210   0.794   0.088
 ...
[613]   4.898 164.920 104.150  33.165 224.779 148.940   4.865 286.847  46.347
[622] 112.430 142.863 119.660  51.981 306.770 230.073 148.949  11.966  46.414
[631] 106.663  10.075  20.619 200.830 634.000
```

# Diamo un’occhiata ai dati

```
> str(BasalMetabolicRate)
Classes ‘tbl_df’, ‘tbl’ and 'data.frame':  667 obs. of  6 variables:
 $ ...1                   : chr  NA "PROTOTHERIA: MONOTREMATA" "Order:
Monotremata" NA ...
 $ species                : chr  NA NA NA "Tachyglossus aculeatus" ...
 $ BasalMetabolicRate_W: num  NA NA NA 2.4 6.78 ...
 $ BodyMass_g             : num  NA NA NA 2725 10300 ...
 $ Body_temp              : chr  NA NA NA "30.7" ...
 $ Ambient_temp           : chr  NA NA NA "19.7" ...
>
```

# 667 osservazioni (i dati riferiti ad altrettante specie) e 6 variabili (in realtà sono 5): specie, tasso metabolico basale (in Watt), massa corporea (in g), temperatura corporea, temperatura ambientale

# Diamo un’occhiata ai dati

```
> BasalMetabolicRate$Body_temp =
as.numeric(BasalMetabolicRate$Body_temp)
Messaggio di avvertimento:
NA introdotti per coercizione

> BasalMetabolicRate$Ambient_temp =
as.numeric(BasalMetabolicRate$Ambient_temp)
Messaggio di avvertimento:
NA introdotti per coercizione
```

# Diamo un’occhiata ai dati

```
> str(BasalMetabolicRate)
Classes ‘tbl_df’, ‘tbl’ and 'data.frame':  667 obs. of  6 variables:
 $ ...1                  : chr  NA "PROTOTHERIA: MONOTREMATA" "Order: 
Monotremata" NA ...
 $ species               : chr  NA NA NA "Tachyglossus aculeatus" ...
 $ BasalMetabolicRate_W: num  NA NA NA 2.4 6.78 ...
 $ BodyMass_g            : num  NA NA NA 2725 10300 ...
 $ Body_temp             : num  NA NA NA 30.7 30.8 32.1 NA NA 35 35 ...
 $ Ambient_temp          : num  NA NA NA 19.7 16.2 12.9 NA NA 19.3 
21.5 ...
>
```

# Diamo un’occhiata ai dati

```
> library(tidyverse)
```

# carichiamo in memoria il pacchetto <`tidyverse`>

```
> str(BasalMetabolicRate)
tibble [667 × 6] (S3: tbl_df/tbl/data.frame)
 $ ...1                   : chr [1:667] NA "PROTOTHERIA: MONOTREMATA"
"Order: Monotremata" NA ...
 $ species                : chr [1:667] NA NA NA "Tachyglossus aculeatus"
...
 $ BasalMetabolicRate_W: num [1:667] NA NA NA 2.4 6.78 ...
 $ BodyMass_g             : num [1:667] NA NA NA 2725 10300 ...
 $ Body_temp              : num [1:667] NA NA NA 30.7 30.8 32.1 NA NA 35
35 ...
 $ Ambient_temp           : num [1:667] NA NA NA 19.7 16.2 12.9 NA NA
19.3 21.5 ...
>
```

# 667 osservazioni (i dati riferiti ad altrettante specie) e 6 variabili (in realtà sono 5): specie, tasso metabolico basale (in Watt), massa corporea (in g), temperatura corporea, temperatura ambientale

**Definizioni**

**Variabile**: una quantità numerica misurabile (carattere biometrico) oppure una proprietà qualitativa (specie, sesso, isola ecc.)

**Valore**: stato della variabile (misura o caratteristica effettiva): 34 mm, 479 mm, maschio, pinguino di Adelia, pigoscelide antartico, Isola Biscoe, Isola Dream ecc.

**Osservazione**: insieme delle rilevazioni effettuate su un singolo soggetto

**Dati tabellari**: insieme di valori, ognuno associato ad un'osservazione (animale) e ad una variabile. I dati tabellari sono «*tidy*» se ogni valore è contenuto in una singola cella, ogni variabile in una colonna ed ogni osservazione in una riga

# La relazione fra peso corporeo e tasso metabolico basale

```
> ggplot(data = BasalMetabolicRate, mapping = aes(x = BodyMass_g, y =
BasalMetabolicRate_W)) + geom_point()

> ggplot(data = BasalMetabolicRate, mapping = aes(x = BodyMass_g, y =
BasalMetabolicRate_W)) + geom_point() + geom_smooth(method = 'loess') +
labs(title = "Body mass and Basal Metabolic Rate",
subtitle = "Scaling of basal metabolic rate with body mass and
temperature in mammals, Clarke et al., 2010",
x = "Body mass (g)", y = "Basal Metabolic Rate (Watt)")


> log10_BodyMass_g = log10(BasalMetabolicRate$BodyMass_g)
> log10_BasalMetabolicRate_W =
log10(BasalMetabolicRate$BasalMetabolicRate_W)
> log10_BasalMetabolicRate = tibble(log10_BodyMass_g,
log10_BasalMetabolicRate_W)
```

# La relazione fra peso corporeo e tasso metabolico basale

```
> log_peso_corporeo = log10_BodyMass_g[!is.na(log10_BodyMass_g)]
> hist(log_peso_corporeo)
```

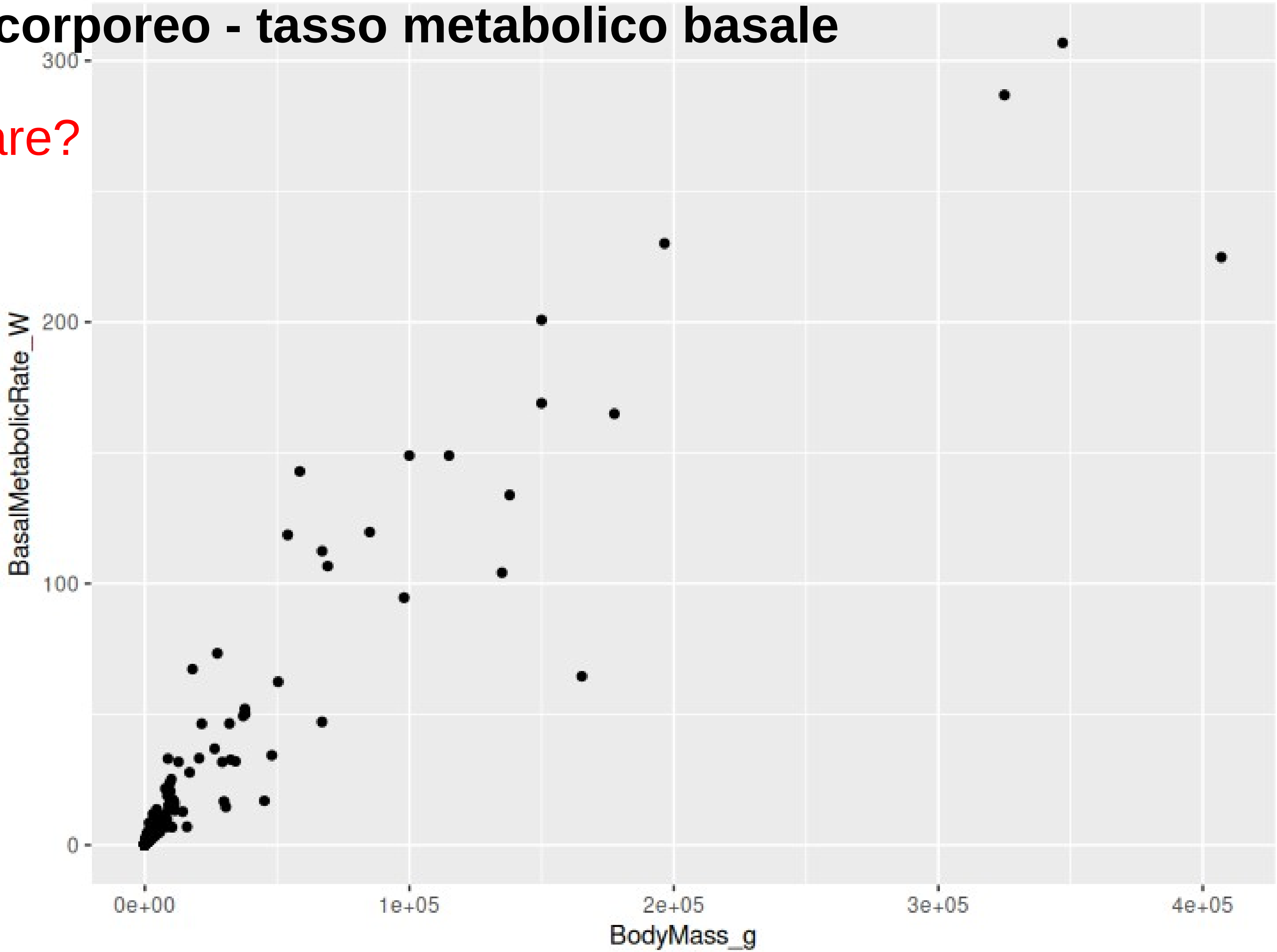
# La relazione fra peso corporeo e tasso metabolico basale

```
> ggplot(data = log10_BasalMetabolicRate, mapping = aes(x =
log10_BodyMass_g, y = log10_BasalMetabolicRate_W)) + geom_point() +
geom_smooth(method = 'lm')

> ggplot(data = log10_BasalMetabolicRate, mapping = aes(x =
log10_BodyMass_g, y = log10_BasalMetabolicRate_W)) +
geom_point() +
geom_smooth(method = 'lm') +
labs(title = "Log10 Body mass and Log10 Basal Metabolic Rate",
subtitle = "Scaling of basal metabolic rate with body mass and
temperature in mammals, Clarke et al., 2010",
x = "Log10 Body mass (g)", y = "Log10 Basal Metabolic Rate (Watt)") +
scale_color_colorblind()
```

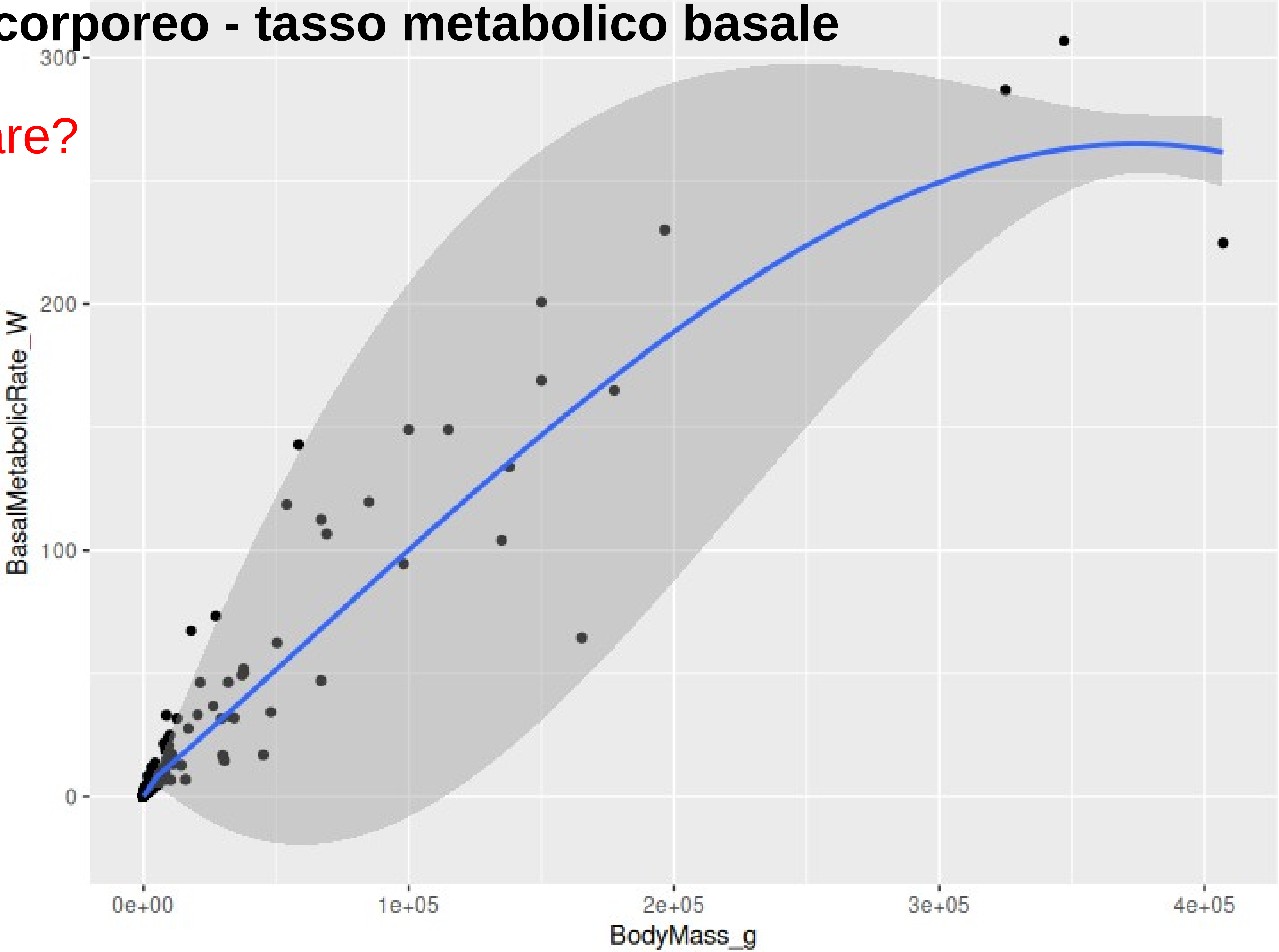
# Scatter plot peso corporeo - tasso metabolico basale

La relazione è lineare?



# Scatter plot peso corporeo - tasso metabolico basale

La relazione è lineare?



PHYSIOLOGICAL REVIEWS

VOL. 27 OCTOBER, 1947 No. 4

# BODY SIZE AND METABOLIC RATE

MAX KLEIBER

*Division of Animal Husbandry, College of Agriculture, University of California, Davis*

CORRELATION BETWEEN BODY SIZE AND METABOLIC RATE. Günther (1944) introduces a recent review on body weight and metabolic rate with a motto which starts as follows:

"It is believed that far greater progress will be made by discarding all thoughts of a uniformity in heat loss and emphasizing the non-uniformity in heat production. . . ."

The sentence is a citation from Benedict's book, *Vital Energetics* (1938, p. 194).

It is rather difficult to understand how forgetting all thoughts of uniformity and emphasizing non-uniformity can stimulate a comparison of metabolic rates of large and small animals. Any comparison presupposes a common basis, and if I were convinced of the "futility of attempts to discover a unifying principle in metabolism" (Benedict, l.c., p. 178) I should not attempt to write a review on the relation of body size and metabolic rate.

The reader can be expected to spend time on this review only when he can be convinced that body size and metabolic rate are actually related. That these two variables are related is in fact common knowledge.

Does a horse produce more heat per day than a rat or do some rats produce more heat than do some horses? Almost anybody who understands what is meant by "heat production per day" will not hesitate to give the correct answer and will even be convinced that the daily rate of heat production of men or sheep is greater than that of rats, but smaller than that of horses. Thus most people (among those who understand the question) are convinced that in general the bigger homeotherms produce more heat per day than the smaller homeotherms, that, in other words, the metabolic rate of homeotherms is positively correlated to body size.

The answer to the next question: "does a horse produce more heat per day per kilogram of body weight than a rat?" requires some biological training. Most

# Basal Metabolic Rate and Body Weight

The positive correlation between metabolic rate and body size, and the **negative correlation between metabolic rate per unit weight and body size**, establish two limits between which we expect to find the rate of heat production of a horse if we know the rate of heat production of a rat.

**We expect the metabolic rate of the horse to be somewhat between that of the rat, and that of the rat times the ratio of horse weight to rat weight**, provided of course that we do not regard these two correlations as simply accidental.

# Basal Metabolic Rate and Body Weight

Bergmann and Leuckart (1855) concluded from measurements of Regnault and Reiset that **the metabolic rate per unit weight was especially great for small animals**.
In four days sparrows consumed as much oxygen as they weighed.
This today would be considered a very high metabolic rate, about four times as high as the rate observed in a sparrow by Benedict (1938).

Richet (1889) discovered “apres coup,” that is empirically, that **the metabolic rate per unit weight of rabbits increased consistently as the body weight decreased**.

**The metabolic rate per unit surface area**, however, **was unaffected by body size**, so Richet concluded that, for metabolic rate, surface area was more important than body weight. Simultaneously Rubner noted a **systematic decrease of the metabolic rate per unit weight of fasting dogs as the weight increased from little dogs of 3 kg to big specimens of 31 kg**.

# The discovery of the surface law

The task is to find a metabolic body size which is chosen so that **the metabolic rate per unit of this body size is the same for large and small animals**.

The **square meter of body surface** is such a unit which allows a more accurate prediction of metabolic rate than the two limits mentioned above.

# The discovery of the surface law

The **metabolic rate** (heat production per unit of time), in particular the basal metabolic rate of humans, is generally expressed in **kilocalories per square meter of body surface**.

This procedure is based on the theory that **in animals of different body size the metabolic rate is proportional to their respective surface areas**. This theory, called the surface law, is now a little over a century old.

| SPECIE | PESO CORPOREO (Kg) | Kcal TOTALI / die | Kcal/Kg DI PESO | Kcal/m² DI SUPERFICIE |
|---|---|---|---|---|
| **Topo selvatico dal dorso striato** (*Apodemus agrarius*) # | 0,0205 | 11,316 | 552 | - |
| colibrì | 0,01 | 2,2 | 220 | - |
| topo | 0,1 | 12,5 | 125 | (1185) |
| cavia | 0,5 | - | - | 1246 |
| faraona | 1 | 70 | 70 | - |
| ovaiola | 2 | 120 | 60 | (943) |
| cane | 10 | 400 | 40 | - |
| cane | 15 | 773 | - | 1039 |
| uomo | 70 | 1700 | 24 | (1042) |
| suino | 100 | 2200 | 22 | (1074) |
| cavallo | 441 | 4983 | 11,3 | 948 |
| bovina da latte | 500 | 7500 | 15 | - |
| toro | 1000 | 12500 | 12,5 | - |
| balena | 10000 | 70000 | 7 | - |

Valori del **Metabolismo Basale** (produzione di calore a riposo in condizioni di neutralità termica) in diverse specie animali per Kg di peso e per m² di superficie corporea. I valori riportati fra parentesi nella quinta colonna non si riferiscono ai dati riferiti alle colonne 2-4. Da Giulio, 1992;
# Andrzej GÓRECKI, Metabolic Rate and Energy Budget of the Striped Field Mouse, *ACTA THERIOLOGICA* VOL. XIV, 14: 181—190. BIAŁOWIEŻA 30.VIII.1969.

# Basal Metabolic Rate and Body Weight

**When the metabolic rate was expressed per square meter of body surface**, however, **the effect of body size disappeared**.
From this and similar observations, Rubner deduced his simple rule that **fasting homeotherms produce daily 1000 kcal of heat per square meter of body surface**.

A 441 kg horse produces over 948 kcal. daily per square meter of body surface, a 64 kg man 1042, a 15 kg dog 1039, and a 2 kg hen 1008.

So well established appeared the surface law that data which did not confirm it were either explained by particular conditions or discarded as results of faulty measurements.

# The “true” body surface area

**Large and small bodies of similar shape have surface areas in proportion to** the squares of their linear dimensions or **the 2/3 power of their volumes**.
If the two bodies have also the same density, then their surface areas are also in proportion to the 2/3 power of their weights.

In **R** (cubo di lato = 10):

```
> superficie.cubo.l.10 = 6*(10^3)^(2/3)
> volume.cubo.l.10 = 10^3
> ratio.sup.vol.cubo.l.10 = superficie.cubo.l.10/volume.cubo.l.10
> ratio.sup.vol.cubo.l.10
[1] 0.6
```

# cubo di lato = 100:

```
> superficie.cubo.l.100 = 6*(100^3)^(2/3)
> volume.cubo.l.100 = 100^3
> ratio.sup.vol.cubo.l.100 = superficie.cubo.l.100/volume.cubo.l.100
> ratio.sup.vol.cubo.l.100
[1] 0.06
>
```

# The “true” body surface area

Many methods have been invented for measuring the surface area of animals.
In their eagerness to refine the surface measurements, many workers in this field seem to have overlooked a major question:

“What is meant by the surface area?”

Unless this question can be answered definitely, how can one decide which method measuresthe surface more accurately?

One should obviously know **whether or not the “true” surface of a rabbit is to include the surface area of the rabbit ears**.

As long as this question is open, which means an uncertainty of about 20 per cent, what is gained by refining the surface measurements to an accuracy of one per cent?

# The “true” body surface area

A great many published results of good work on metabolic rates are practically lost for comparative physiology because they are expressed only per unit of surface area, and the authors did not furnish the data which would make a comparison with other work possible.

In the Annual Review of Physiology (Kleiber, 1944) alarm was again expressed as follows:

“*In 10 papers (from 8 laboratories) studied for this review metabolic rates of rats are expressed per unit of the surface area. Four of the 10 authors did not state how they measured or calculated this area. One multiplied the 2/3 power of body weight (in kg) by 7.42, another by 9.1, a third by 10, to calculate the surface area in square decimeters*”.

# Theoretical validity of the surface law

The theories advanced for the interpretation of the surface law of animal metabolism may be classified into 5 major groups:
**The metabolic rate of animals must be in proportion to their body surface**.

1. **The rate of heat transfer between animal and environment is proportional to the body surface area**.

2. The intensity of flow of nutrients, in particular oxidizable material and oxygen, is a function of the sum of internal surfaces which in turn is proportional to the body surface.

3. The rate of supply of oxidizable material and oxygen to the tissues is a function of the mean intensity of the blood current, which is proportional to the square area of the blood vessels, which in turn is proportional to the area of body surface.

# Theoretical validity of the surface law

4. The composition of the animals is a function of their body size.
the larger the animal the lower is the ratio of the mass of metabolically active organs to the mass of metabolically inert organs.

5. The cells of the body have an inherent requirement of oxygen consumption per unit weight, which is smaller the larger the animal.

# Theoretical validity of the surface law

Il flusso di calore di un corpo rivestito da uno strato isolante è dato dall’equazione di Fourier:

$$q = S\,\lambda\,\frac{T_i - T_s}{L}$$

*q: quantità di calore dispersa nell'unità di tempo*
*S: area della superficie corporea*
*L: spessore del rivestimento (pannicolo adiposo sottocutaneo + cute + pelliccia*
$T_i$*: temperatura corporea interna*
$T_s$*: temperatura della superficie corporea*
$\lambda$*: conducibilità termica*

# Theoretical validity of the surface law

For a given difference between internal temperature and surface temperature, the rate of heat transfer is proportional to the surface area when the specific insulation for large and small bodies is the same.

The specific insulation of animals is, however, variable.

The classical demonstration of this fact is the experiment of Hoesslin (1888). He reared two littermate dogs, one at 32°C, the other at 5°C. The latter had to cope with a temperature difference between body and environment six times as great as the corresponding difference for his brother. Yet the metabolic rate of the dog in the cold was only 12 % higher. **He solved the problem of keeping warm by growing a fur that weighed three times as much as that of his brother**.

# Theoretical validity of the surface law

Instead of maintaining the metabolic rate per unit surface area constant, large and small animals therefore might maintain a constant metabolic rate per unit weight, and with a **variable specific insulation** adapt the rate of heat loss to that metabolic rate.

Se un topo del peso di 60 g avesse lo stesso tasso metabolico basale di un bue per unità di peso, per mantenere una T corporea costante in un ambiente a 3°C dovrebbe avere un rivestimento superficiale spesso 20 cm!

This rather extreme example illustrates why it is advantageous for a small animal to have a higher metabolic rate per unit weight than a large animal.

# Theoretical validity of the surface law

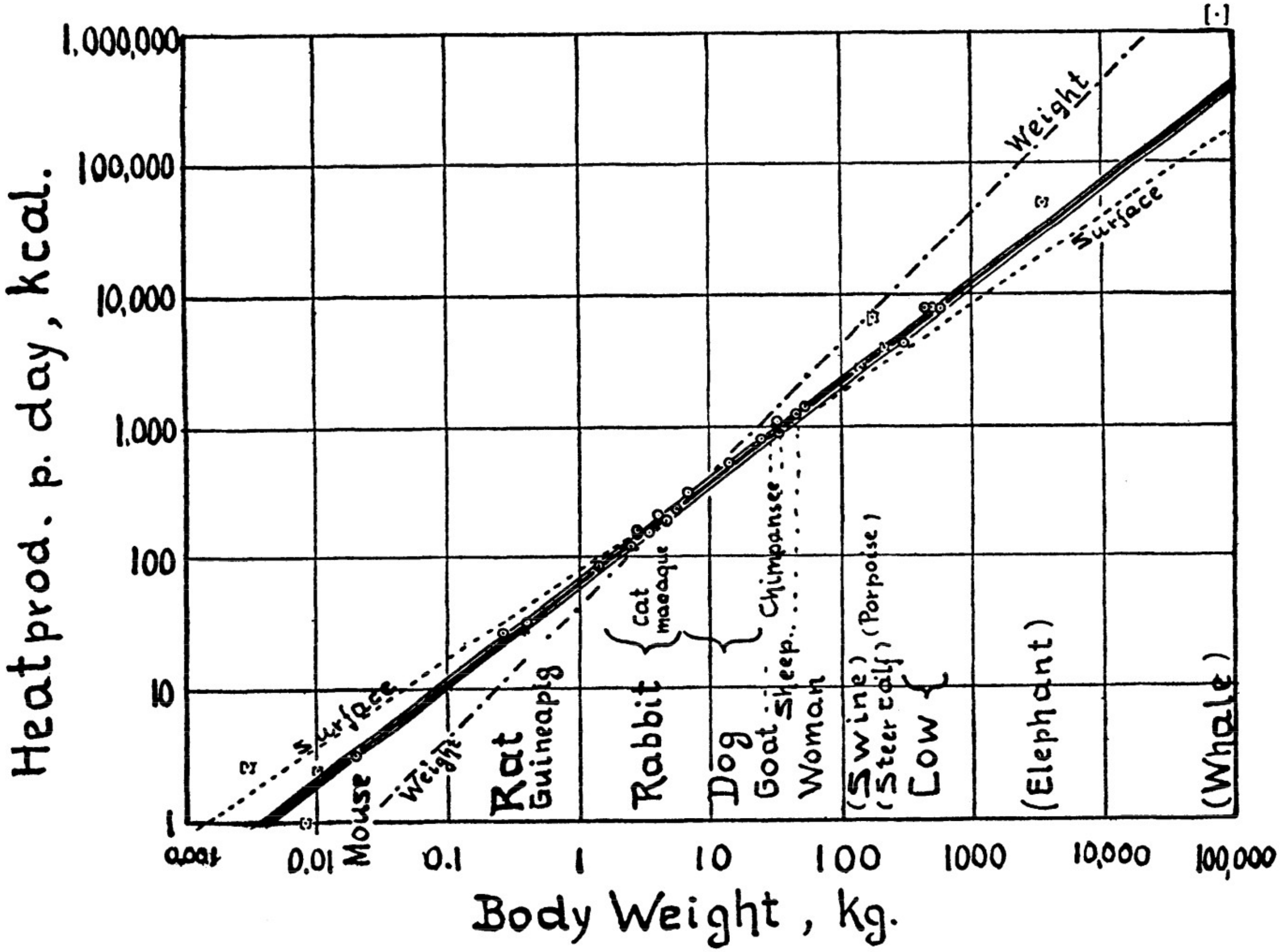
In natural selection, those animals probably prove to be the fittest whose cells are adapted to such a level of oxygen consumption that the metabolic rate of the animal is most suitable for the maintenance of a constant body temperature and in line with the most efficient transport of oxygen.

A four ton animal, whose cells insisted on a rate of oxygen consumption per unit weight equal to that of mouse cells, could not survive, because such a metabolic rate could not be supported by the circulatory system and would upset the maintenance of a constant body temperature (andrebbe in ipertermia).

## Taking the logarithms of basal metabolic rate and body weight produces a linear mathematical relationship

The deviations of the empirical results from the surface law justified the **search for a function of body size to which metabolic rate might be more nearly proportional than to body surface**

Plotting the logarithms of fasting metabolic rate against the logarithms of body weight revealed a linear relation between these two variables, with surprisingly small deviations from the mean trend.



# Taking the logarithms of basal metabolic rate and body weight produces a linear mathematical relationship

This chart shows the logarithms of the mean metabolic rates for 28 groups of animals plotted against the logarithms of the corresponding mean body weights.

The animals considered ranged in size from 20 gram mice to nearly 4 ton elephants.

A **regression line** indicates the average trend, and Benedict notes "a most gratifying straight line relationship between the total heat production and the body weight."

**Taking the logarithms of basal metabolic rate and body weight produces a linear mathematical relationship**

The surface area was not well enough defined to serve as a basis for measurement, and a power function of body weight was suggested as the basis of metabolic body size.

**When the logarithm of metabolic rate is a linear function of the logarithm of body weight**, then **metabolic rate is proportional to a given power of body weight**.

The metabolic rate was more nearly proportional to the 3/4 power of body weight than to either the 2/3 power or the surface area of the animals.

**The 3/4 power was proposed as the best fitting function** (Kleiber, 1932).

# Il modello finale

In **R**:

```
> library(tidyverse)
> str(BasalMetabolicRate)
tibble [667 × 6] (S3: tbl_df/tbl/data.frame)
 $ ...1                  : chr [1:667] NA "PROTOTHERIA: MONOTREMATA" "Order:
Monotremata" NA ...
 $ species               : chr [1:667] NA NA NA "Tachyglossus aculeatus" ...
 $ BasalMetabolicRate_W: num [1:667] NA NA NA 2.4 6.78 ...
 $ BodyMass_g            : num [1:667] NA NA NA 2725 10300 ...
 $ Body_temp             : chr [1:667] NA NA NA "30.7" ...
 $ Ambient_temp          : chr [1:667] NA NA NA "19.7" ...
> MetabolicWeight =  BasalMetabolicRate$BodyMass_g^0.75
> MetabolicWeight[1:10]
 [1]         NA         NA         NA  377.15935 1022.41666  135.06721
 [7]         NA         NA   77.24972  169.08641
>
```

# Il modello finale

In **R**:

```
> MetabolicWeight =  BasalMetabolicRate$BodyMass_g^0.75
> MetabolicWeight[1:10]
 [1]         NA         NA         NA  377.15935 1022.41666  135.06721
 [7]         NA         NA   77.24972  169.08641
> MetabolicWeight =  ((BasalMetabolicRate$BodyMass_g^3)^0.5)^0.5
> MetabolicWeight[1:10]
 [1]         NA         NA         NA  377.15935 1022.41666  135.06721
 [7]         NA         NA   77.24972  169.08641
>
```

Perché il risultato è il medesimo?

# Il modello finale

Kleiber notò che **se il Metabolismo Basale (MB) viene diviso per il peso corporeo elevato a 3/4, il risultato che si ottiene è costante**, a prescindere dalla specie animale considerata

Questo risultato gli suggerì che la relazione matematica fra il peso corporeo (Body Weight - BW) ed il MB poteva essere la seguente:

$$MB = cost * BW^{0.75}$$

$$\mathbf{MB / BW^{0.75} = cost}$$

«*The unit of the 3/4 power of body weight, kg^3/4, is therefore a suitable unit of metabolic body size*»

# Il modello finale

In **R**:

```
> perUnitMWBasalMetabolicRate =
BasalMetabolicRate$BasalMetabolicRate_W/MetabolicWeight

> perUnitMWBasalMetabolicRate[1:20]
 [1]          NA          NA          NA 0.006373964 0.006629391 0.008010827
 [7]          NA          NA 0.016246013 0.015075132 0.015972209 0.013174186
[13] 0.015482759 0.014890222 0.014901070 0.014577115 0.012415577 0.010286551
[19] 0.013146548          NA
>
```

# Il modello finale

$$y = a\, x^b$$

$\boldsymbol{y}$ = Metabolismo Basale;
$\boldsymbol{a}$ = costante di proporzionalità il cui valore dipende dalla specie di appartenenza;
$\boldsymbol{x}$ = massa corporea;
in base a misure effettuate da Kleiber (1967) su 12 specie di Mammiferi,
$\boldsymbol{b} \cong 0{,}75$;
$\boldsymbol{a} = \mathbf{67{,}6}$.

Allora, possiamo scrivere la seguente equazione, valida in generale:

MB (Kcal) = 67,6 * P $^{0,75}$ (Kg)
MB (Mcal) = 0,0676 * P $^{0,75}$ (Kg)

$$\log y = \log (a\, x^b)$$
$$\log y = \log a + b \log x$$
$$\log M.B.\ (Kcal) = \log 67{,}6 + 0{,}75 \log x$$

# Il modello finale

The unit of metabolic body size is useful in evaluating levels of food intake in animal production, and in classifying farm animals with regard to their efficiency as food utilizers. Food requirements and dosages of most vitamins and drugs may be expressed in terms of metabolic body size.

# Come misurare il Metabolismo Basale?

BIOENERGETICS IN THE KILLER WHALE, ORCINUS ORCA

by

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Honours B.Sc. Trent University, 1983

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF

THE REQUIREMENTS FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

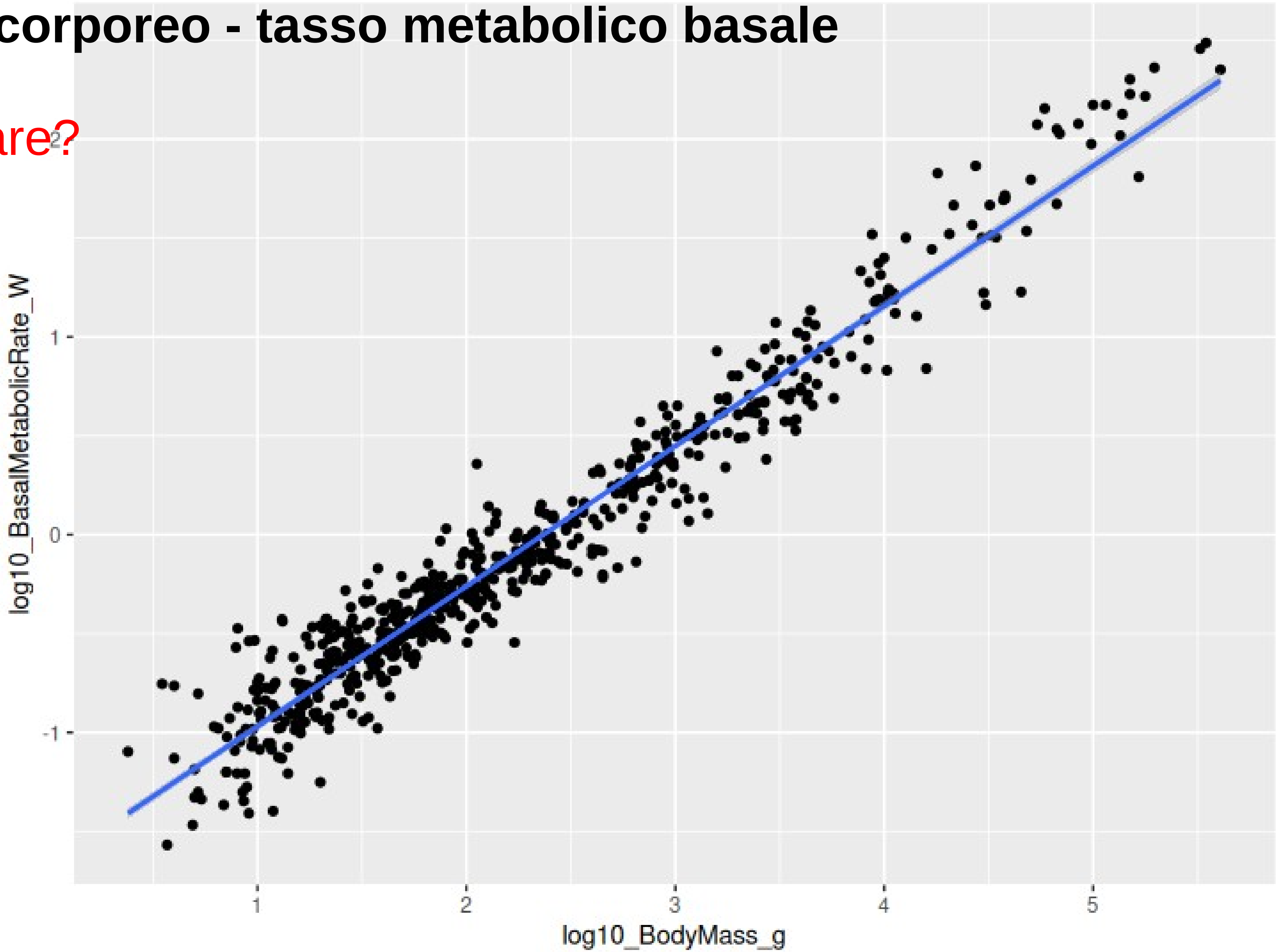
in

THE FACULTY OF GRADUATE STUDIES
DEPARTMENT OF ANIMAL SCIENCE

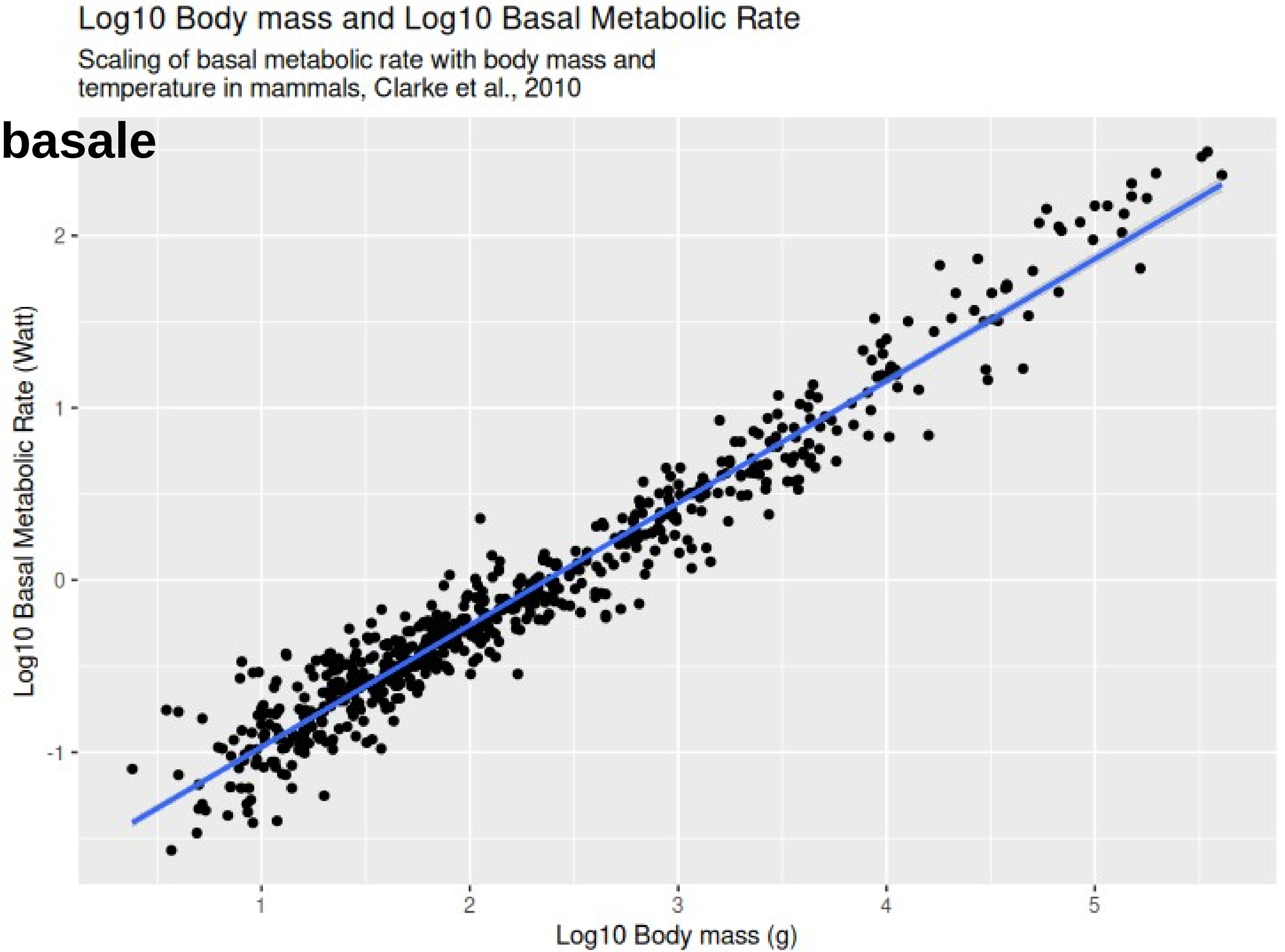
We accept this thesis as conforming
to the required standard

# Scatter plot peso corporeo - tasso metabolico basale

La relazione è lineare?



**Scatter plot**
**peso corporeo -**
**tasso metabolico basale**



# Il modello finale

```
> metabolic_rate_model = log10_BasalMetabolicRate %>%
    lm(log10_BasalMetabolicRate_W ~ log10_BodyMass_g, data = .)

> metabolic_rate_model

Call:
lm(formula = log10_BasalMetabolicRate_W ~ log10_BodyMass_g, data = .)

Coefficients:
     (Intercept)  log10_BodyMass_g
         -1.6783            0.7087
```

Cos'è l'intercetta?

Qual è il coefficiente angolare?

# Il modello finale

```
> log10x = seq(0, 5.7, 0.1)
```

# definiamo la variabile indipendente log10x

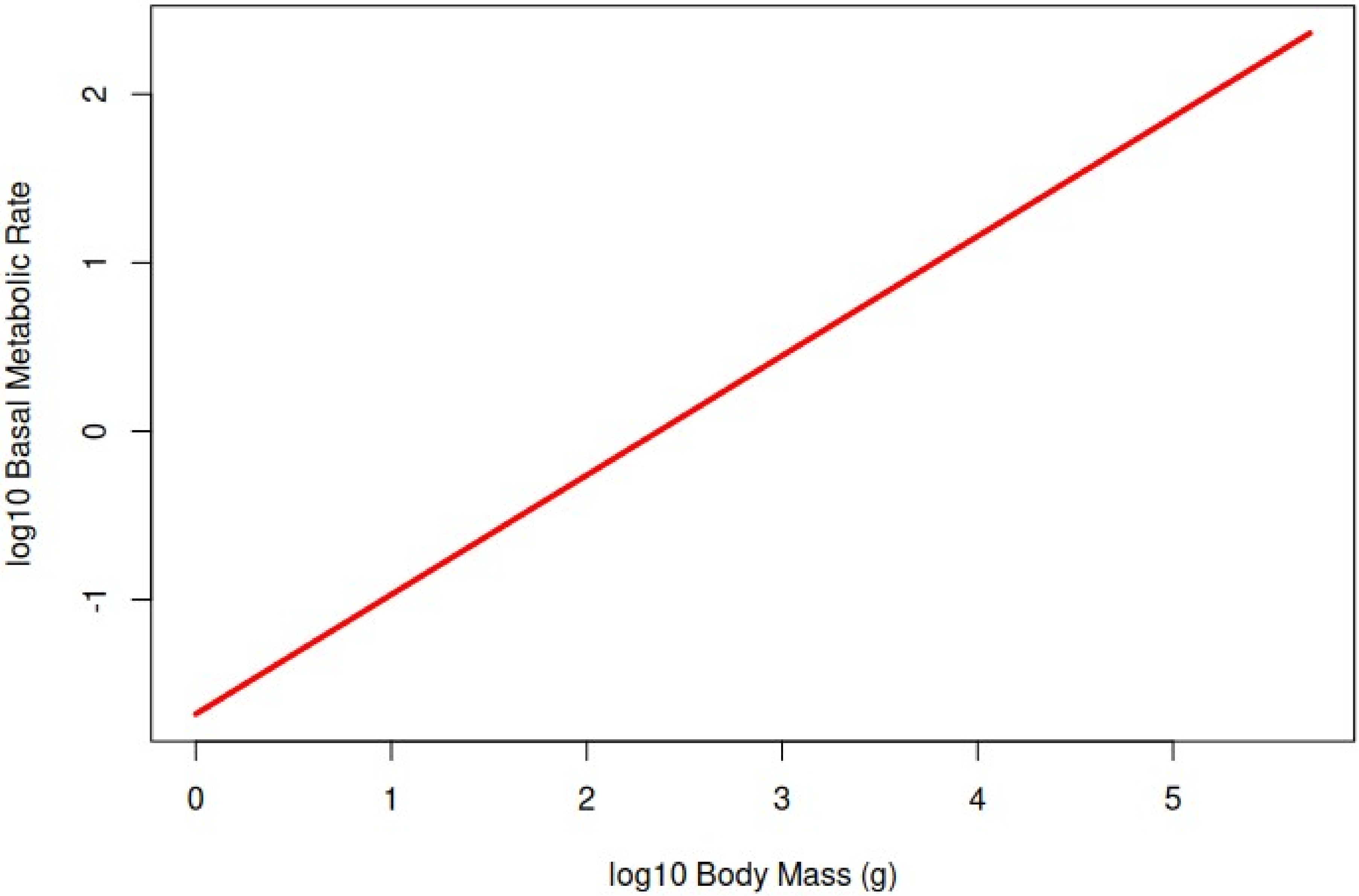
```
> log10x
 [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5
1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 3.0 3.1 3.2
[34] 3.3 3.4 3.5 3.6 3.7 3.8 3.9 4.0 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8
4.9 5.0 5.1 5.2 5.3 5.4 5.5 5.6 5.7
> log10y = -1.6783 + 0.7087*log10x
```

# scriviamo l’equazione

Cosa rappresentano le due variabili?

```
> plot(log10y~log10x, type = 'l', col = 'red', lwd =3, xlab = 'log10
Body Mass (g)', ylab = 'log10 Basal Metabolic Rate')
```

# Il modello finale



# Il modello finale

```
log10y = -1.6783 + 0.7087*log10x

log10y = log10(a) + b*log10x

log10(a) = -1.6783 → a = 10^(-1.6783) = 0.021
> 10^-1.6783
[1] 0.0209749

y = a*x^b
```

$y = 0{,}021 * x^{0{,}7}$

$y$ = tasso metabolico basale (Watt)

$x$ = peso corporeo (g)

Per es., in un suino di 135 kg il Tasso Metabolico Basale sarebbe di ≈ 91 Watt

```
> 0.021*135000^0.7087
[1] 90.80021
```